RAFDAN

رفدان

TEAM / SaudiConsulters

Know what to review before the course starts.

Rafdan is a pre-semester prerequisite-readiness platform that identifies which concepts from a previous course a student actually needs for the upcoming course, diagnoses weak points, and builds a focused review plan.

> **The goal is not to re-teach the whole prerequisite course. It is to find the bridge knowledge the next course depends on - then help the student repair only the gaps that matter.**

Hackathon	Track	Primary user
Farq Hackathon	Student Journey	University student

Why the name still fits: Rafdan comes from the Arabic root rafd - support and reinforcement. Here, Rafdan reinforces the prerequisite knowledge a student needs before moving forward.

Passing a prerequisite course does not mean the knowledge is still available when the next course begins.

Students often enter a new course months after completing its prerequisite. They may have passed the earlier course, but forgotten specific concepts that the new course assumes they already know.

Student problem	Instructor problem
The student does not know what to review. Re-reading an entire old course is inefficient, while reviewing nothing can create difficulty from the first weeks.	The instructor must begin with an assumed baseline. Students can arrive with very different levels of retained prerequisite knowledge.

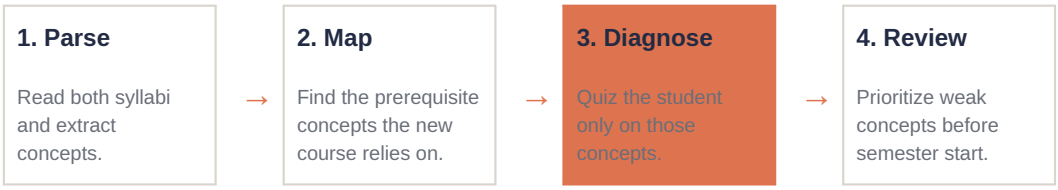
The important insight

Not every topic from the prerequisite course matters equally to the new course. Rafdan first understands the dependency between the two syllabi, then assesses only the prerequisite concepts that are actually relevant.

Generic approach	Rafdan approach
Review the whole old course or take a generic quiz.	Identify bridge concepts between the old and new courses, test those concepts, and prioritize only the gaps that matter.

A dependency-aware diagnostic before the course begins.

The student selects the upcoming course and its prerequisite. Rafdan uses the two syllabi to identify the prerequisite concepts that the upcoming course actually depends on.



Illustrative example

Prerequisite course	Upcoming course	Bridge concepts
Programming I	Data Structures	Arrays, functions, object references, recursion fundamentals, and basic debugging.

The quiz is targeted, not a final exam for the prerequisite course. It answers: Which missing prior concepts are likely to create difficulty in this specific upcoming course?

AI can interpret the syllabi, but the diagnosis should remain structured and auditable.

INPUTS	PROCESSING	OUTPUTS
• Prerequisite syllabus • Upcoming-course syllabus • Known prerequisite relationship • Optional course material	• Concept extraction • Concept normalization • Dependency mapping • Quiz generation • Concept-level scoring	• Readiness by concept • Critical knowledge gaps • Focused review plan • Optional re-check

Suggested role of AI

• Syllabus understanding: extract topics, learning outcomes, terminology, and conceptual relationships.

• Dependency mapping: estimate which prerequisite concepts are directly needed for topics in the upcoming course.

• Question drafting: generate diagnostic questions grounded in the identified concepts and available course materials.

• Explanation: explain why a weak concept matters for the upcoming course and what to review.

What should not be left to an unconstrained LLM

• Final mastery scoring: use explicit concept-level scoring rules.

• Coverage: ensure the quiz covers the intended dependency map instead of whatever the model happens to ask.

• Syllabus facts: do not invent concepts or prerequisites that are not supported by the supplied material.

• High-stakes interpretation: a weak diagnostic result should guide review, not be treated as a formal academic judgment.

The output should tell the student exactly what to review - and why.

A useful result is more specific than a single readiness percentage. It should show concept-level status and connect each gap to the upcoming course.

Concept	Diagnostic	Why it matters next	Action
Arrays	Ready	Used throughout early data-structure implementations.	No review needed
Functions	Review	Required to follow modular implementations and helper methods.	20-min refresher
Object references	Critical gap	Important for linked structures and object relationships.	Review + practice
Recursion basics	Critical gap	Needed later for recursive structures and algorithms.	Review + mini-quiz

Illustrative review plan

Priority 1: Object references - review the relevant section, solve 3 short exercises, then re-test.

Priority 2: Recursion fundamentals - review base case / recursive step, complete one tracing exercise, then re-test.

Priority 3: Functions - quick refresher only; current weakness is smaller and lower-risk.

Rafdan should prioritize review by dependency importance x demonstrated weakness, not simply by the number of questions answered incorrectly.

Prove the core loop on a small number of course pairs.

The MVP does not need to support an entire university catalog. It needs to prove that Rafdan can meaningfully connect two syllabi, diagnose retained prerequisite knowledge, and generate a focused pre-course review plan.

MVP should include

- 1-2 prerequisite course pairs
- Syllabus upload or preloaded syllabi
- Concept-dependency map
- 15-20 question diagnostic
- Concept-level results
- Prioritized review plan
- Optional quick re-test

Do not overbuild yet

- Full LMS integration
- Every university course
- Automated academic advising
- Formal prerequisite enforcement
- Large content marketplace
- Long-term student analytics
- Instructor administration suite

Recommended demo sequence

- Select an upcoming course and its prerequisite.
- Show both syllabi and the generated concept-dependency map.
- Complete a short diagnostic as a student.
- Reveal concept-level strengths and weaknesses.
- Show that the highest-priority gap is important because a specific upcoming topic depends on it.
- Generate a short review plan.
- Re-test one concept to show the readiness loop.

Rafdan becomes stronger if it proves it is more than an AI-generated quiz.

Generic pre-course quiz	Rafdan
Tests broad prior-course knowledge.	Tests prerequisite concepts selected because the new syllabus depends on them.
Incorrect answers become a score.	Incorrect answers become concept-level gaps connected to future course topics.
Same review for everyone.	Personalized review based on the student's actual weaknesses.
Often isolated from curriculum structure.	Uses the explicit relationship between prerequisite and upcoming courses.

Key assumptions to pressure-test

• Can an AI map dependencies between two real syllabi accurately enough to be trusted?

• Should instructors validate the dependency map before students use it?

• How many questions are needed to estimate concept readiness without making the diagnostic too long?

• How should question difficulty affect the mastery estimate?

• Should the review plan point only to topics, or also to university-approved resources?

• Does Rafdan create enough value for students to voluntarily complete it before semester start?

• What measurable outcome best proves impact: quiz improvement, early-course performance, student confidence, or another metric?

Potential long-term direction

If validated, Rafdan could later integrate with the university LMS and automatically invite students to complete a readiness check before courses with prerequisites. Aggregate, privacy-safe results could also help instructors see which foundational concepts need a brief refresher at the beginning of the semester.

Current direction: Rafdan is the selected concept for the hackathon. The immediate priority is validating the dependency-mapping logic, diagnostic design, and MVP scope before expanding the platform.